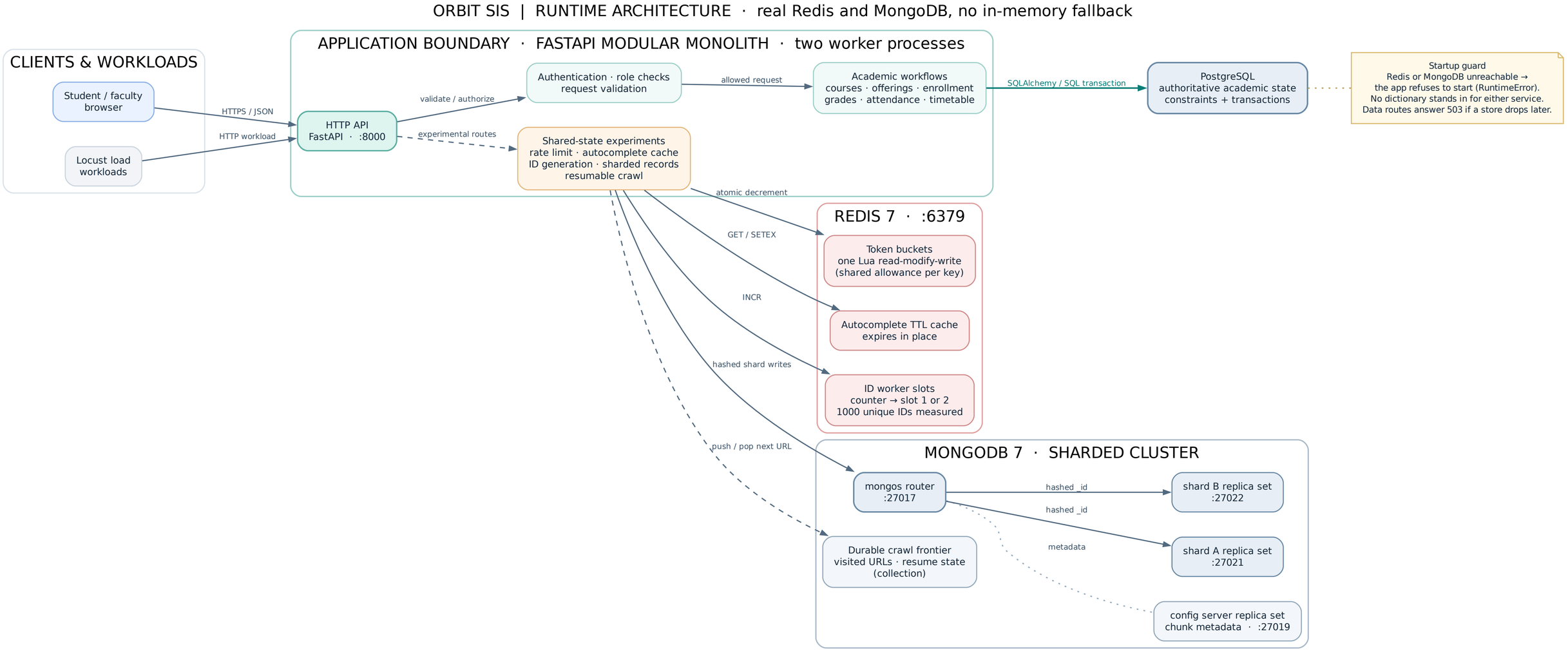
ORBIT SIS | RUNTIME ARCHITECTURE · real Redis and MongoDB, no in-memory fallback
CLIENTS & WORKLOADS
Student / faculty browser
Locust load workloads
HTTPS / JSON
HTTP workload
APPLICATION BOUNDARY · FASTAPI MODULAR MONOLITH · two worker processes
HTTP API FastAPI · :8000
validate / authorize
Authentication · role checks request validation
allowed request
Academic workflows courses · offerings · enrollment grades · attendance · timetable
SQLAlchemy / SQL transaction
PostgreSQL authoritative academic state constraints + transactions
Startup guard
Redis or MongoDB unreachable → the app refuses to start (RuntimeError).
No dictionary stands in for either service.
Data routes answer 503 if a store drops later.
experimental routes
Shared-state experiments rate limit · autocomplete cache ID generation · sharded records resumable crawl
atomic decrement
REDIS 7 · :6379
GET / SETEX
Token buckets one Lua read-modify-write (shared allowance per key)
INCR
Autocomplete TTL cache expires in place
hashed shard writes
ID worker slots counter → slot 1 or 2 1000 unique IDs measured
push / pop next URL
MONGODB 7 · SHARDED CLUSTER
mongos router :27017
hashed _id
shard B replica set :27022
hashed _id
shard A replica set :27021
metadata
config server replica set chunk metadata · :27019
Durable crawl frontier visited URLs · resume state (collection)